Problem 1 | Problems T4_PhaseDiagrams I.pdf #1 | Lever rule · two equations

Phase Limits from the Phase Fractions of Two Alloys

Original statement · Problems T4_PhaseDiagrams I.pdf

1. In the phase diagram of two hypothetic metals, A and B, there is a phase α rich in A and a phase β rich in B. Given the mass fraction of two alloys for a specific temperature (table), determine the composition of the phase limits (or solid solubility limits) of the α and β phases at such temperature.

Alloy composition	*Fraction of α phase*	*Fraction of β phase*
30% B	*0.78*	*0.22*
65% B	*0.36*	*0.64*

Official Solution: $C_\alpha = 11.7\%$, $C_\beta = 95\%$

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium:** both alloys are at the same temperature, in equilibrium, and both contain α and β only, so both lie in the α + β two-phase region [Session 7 Slide 26].
2. **One tie line:** at constant pressure a two-component, two-phase system has $F = C - P + 1 = 2 - 2 + 1 = 1$ degree of freedom, and fixing the temperature uses it [Session 7 Slide 13]. The compositions of α and β are therefore the same for both alloys: C_α and C_β are the two ends of one tie line.
3. **Mass fractions** obey the lever rule [Session 7 Slide 16].

Data

Quantity	Alloy 1	Alloy 2
Overall composition C_0 (% B)	30	65
W_α	0.78	0.36
W_β	0.22	0.64

Unknowns: C_α and C_β (% B). Two unknowns and two independent equations (one per alloy; $W_\beta = 1 - W_\alpha$ adds nothing new).

PHASE 2 · FRAMES & STRATEGY

The only frame is the phase diagram itself: composition in % B along the horizontal axis and temperature along the vertical axis. All the work is done on one horizontal line (the tie line at the given temperature), so it is one-dimensional and no change of basis is involved.

Why the lever rule. It is the tool that relates the amounts of the phases to the positions of the alloy and of the phase limits on the tie line [Session 7 Slide 16]:

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha}$$

Writing it once for each alloy gives two linear equations in C_α and C_β.

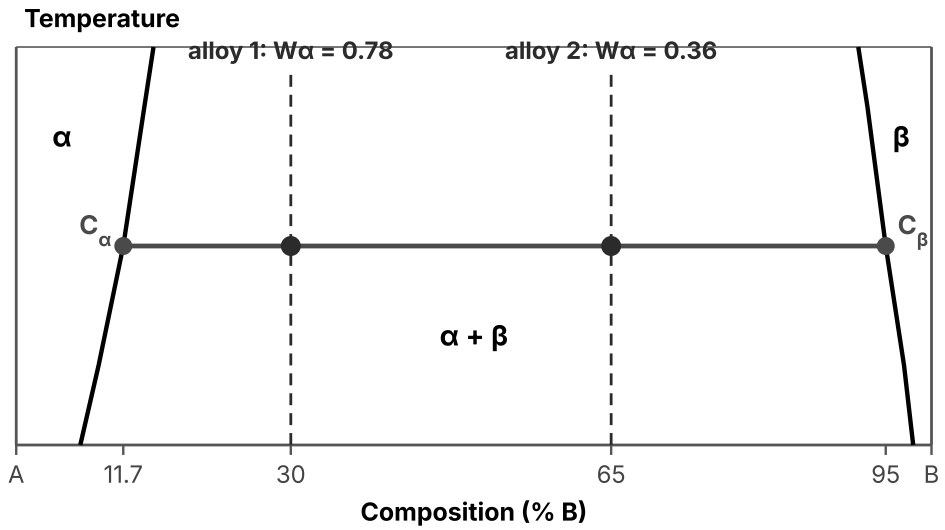


Figure 1. Both alloys are in the α + β region at the same temperature, so they share the tie line from C_α to C_β; the two lever-rule equations give C_α = 11.7 % B and C_β = 95 % B.

PHASE 3 · STEP-BY-STEP DERIVATION

1. The two equations Call $D = C_\beta - C_\alpha$ the length of the tie line. Multiplying the lever rule by D:

$$W_\alpha D = C_\beta - C_0$$

Alloy 1 ($C_0 = 30, W_\alpha = 0.78$) and alloy 2 ($C_0 = 65, W_\alpha = 0.36$):

$$0.78\,D = C_\beta - 30 \qquad (1)$$

$$0.36\,D = C_\beta - 65 \qquad (2)$$

2. Length of the tie line Subtracting (2) from (1) eliminates C_β:

$$0.78\,D - 0.36\,D = (C_\beta - 30) - (C_\beta - 65) \;\Rightarrow\; 0.42\,D = 35 \;\Rightarrow\; D = \frac{35}{0.42} = 83.33\ \%\mathrm{B}$$

3. Limit of β From (1):

$$C_\beta = 30 + 0.78 \times 83.33 = 30 + 65.00 = 95.0\ \%\mathrm{B}$$

4. Limit of α

$$C_\alpha = C_\beta - D = 95.0 - 83.33 = 11.67 \approx 11.7\ \%\mathrm{B}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **β fractions** (not used in the solution): $W_\beta = \dfrac{C_0 - C_\alpha}{C_\beta - C_\alpha}$ gives $\dfrac{30 - 11.67}{83.33} = 0.22$ and $\dfrac{65 - 11.67}{83.33} = 0.64$, exactly the values of the table.
- **Mass balance of B** for alloy 1: $0.78 \times 11.67 + 0.22 \times 95.0 = 9.10 + 20.90 = 30.0\ \%\mathrm{B} = C_0$.
- **Order:** $C_\alpha = 11.7 < 30 < 65 < 95 = C_\beta$, so both alloys are indeed inside the α + β region, as assumed.
- **Units:** all compositions are in % B (mass), fractions are dimensionless; each equation is homogeneous in % B.

- **Limiting cases** (fractions dimensionless, compositions in the units of the statement): with the limits found, an alloy at $C_0 = C_\alpha$ would be all α ($W_\alpha = 1$) and one at $C_0 = C_\beta$ all β ($W_\alpha = 0$); both given alloys lie between the two limits, as fractions between 0 and 1 require.

Interpretation At this temperature α can dissolve at most 11.7 % B and β at most 5 % A (95 % B). The alloy closer to the α limit (30 % B) has more α, as the lever rule predicts.

Check against the official solution

The official solution gives $C_\alpha = 11.7\%$ and $C_\beta = 95\%$; the result agrees. The official solutions PDF has no worked page for this problem; only the key of the statement is available.